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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech (CST) (CBCS) Degree Examinations
 December 2024- January 2025
22CST101- MECHANICS OF DEFORMABLE BODIES

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each unit

		Unit – I												
		Marks	BT*	CO*	PO*									
1.	Derive Stress Strain relationship in 3 D Stress System.	20	L*2	1	1,2,3									
2.	A homogeneous isotropic elastic solid has the displacement field given by: $u = (x^3y^2z - 4xy^2z^2) 10^{-4}$ $v = (x^2y^2z^3 + 3xyz^2) 10^{-4}$ $w = (2x^4 - 3y^2z^2 + xyz^4) 10^{-4}$ Take $E = 200\text{GPa}$ and $\mu = 0.3$ Evaluate the stress component at the point (1,2,3)	20	L3	1	1,2,3									
		Unit – II												
3.	The state of stress at a point is given by <table border="1" style="display: inline-table; vertical-align: middle;"><tr><td>-8</td><td>4</td><td>5</td></tr><tr><td>4</td><td>12</td><td>-6</td></tr><tr><td>5</td><td>-6</td><td>-4</td></tr></table> MPa Determine the stresses on a plane whose normal has DC's $l=1/4$, $m = 1/2$ and also find the normal and shear stresses on that plane.	-8	4	5	4	12	-6	5	-6	-4	20	L3	2	1,2,3
-8	4	5												
4	12	-6												
5	-6	-4												
4.	The strain tensor at a point is given by: <table style="display: inline-table; vertical-align: middle;"><tr><td>8×10^{-4}</td><td>1×10^{-4}</td><td>4×10^{-4}</td></tr><tr><td>1×10^{-4}</td><td>4×10^{-4}</td><td>5×10^{-4}</td></tr><tr><td>4×10^{-4}</td><td>5×10^{-4}</td><td>7×10^{-4}</td></tr></table> Calculate principle strains and their directions.	8×10^{-4}	1×10^{-4}	4×10^{-4}	1×10^{-4}	4×10^{-4}	5×10^{-4}	4×10^{-4}	5×10^{-4}	7×10^{-4}	20	L3	2	1,2,3
8×10^{-4}	1×10^{-4}	4×10^{-4}												
1×10^{-4}	4×10^{-4}	5×10^{-4}												
4×10^{-4}	5×10^{-4}	7×10^{-4}												
		Unit – III												
5.	a) Deduce that function $\varphi = A \log r + Br^2 \log r + Cr^2 + D$. Solve the problem of axis-symmetric stress distribution.	10	L3	3	1,2									
	b) Given $\varphi = (p/r) \cos \theta$, where P is a constant . Show that φ is a airy's stress function.	10	L3	3	1,2									
6.	Investigate what problem of plane stress can be solved by the stress function $\phi = 3F/4h (xy - xy^3/3h^2) + py^2/2$ where F and p are constants. Apply to the region included in $y=0$, $y=\pm h$, $x=0$ on the side x is positive.	20	L3	3	1,2,3									
		Unit – IV												
7.	Derive the relationship between Torque and Shear Flow and Twist and Shear Flow.	20	L2	4	1,2									
8.	Consider a solid circular section of 100mm diameter. Determine the thickness of rectangular tube section so that both sections have the same maximum shear stress (ζ) for the given torque (T) and both have same stiffness (Twist / unit length) Θ	20	L3	4	1,2,3									
		Unit-V												
9.	Explain any three theories of failure	20	L3	5	1,2,3									
10.	Draw stress strain curve for different materials.	20	L2	5	1,2									
BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome														

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